Prediction Is Not Fact

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Prediction Is Not Fact before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Boundaries and 'Not Sure'

Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Build a Human Classifier

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Repeating Efficiently

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Prediction Is Not Fact trace and record the first step you can explain without help.

Use W2: Check a boundary without changing the rule for **Prediction**s and **observation**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Prediction Is Not Fact explanation against the supplied facts before discussing it.

Use the worked case for **Prediction**s and **observation**s. Proposed explanation: “Running the rule correctly proves the **prediction** correct”. Decide whether this explanation is supported.

Point to the step or supplied **fact** that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Transfer the check to drying predictions

In a fictional paper test, Dry1 predicts slow for a moisture score of at least 7, quick otherwise. Allowed scores are whole numbers 1 through 10.

The observed outcome is slow if drying time is greater than 10 minutes, quick otherwise. Records are P: score 6, time 12; Q: score 7, time 10; R: score 9, time 14; S: score 5, time missing.

Record all **prediction**s, observed categories and comparison statuses. Summarise matches out of assessed cases, name the **unassessed** count, and limit the conclusion.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Prediction Is Not Fact target to a trace and a justified conclusion.

On Monday a team predicts 'rain on Tuesday'. On Tuesday they count a damp playground as success, although sprinklers ran. Write an **observation** rule that could have been agreed before Tuesday.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Prediction Is Not Fact case and show the deciding evidence.

A weather helper predicts tomorrow's rain.

> Is this a **fact** about tomorrow? How can it be checked?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A model predicts rain tomorrow. Tomorrow is dry. Was the **prediction** an **observation** of tomorrow? Explain what new **evidence** settles the result.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 17 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Prediction Is Not Fact adds to Helpful Classroom Sorter and keep the evidence.

Label the sorter's suggested bin as a **prediction** until a human confirms the final placement.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

On Monday a team predicts 'rain on Tuesday'. On Tuesday they count a damp playground as success, although sprinklers ran. Write an **observation** rule that could have been agreed before Tuesday.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

W1 Separate inputs from observed outcomes

> Separate inputs from observed outcomes. Apply the supplied Prediction Is Not Fact facts and keep your reasoning.

For Ramp1 trial B, height is 6 cm and the later travel distance is 3 units. State the **input**, **prediction**, **observation**, observed category and comparison status.

Explain why the **prediction** rule and **outcome rule** use different measurements.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Check a boundary without changing the rule

> Check a boundary without changing the rule. Apply the supplied Prediction Is Not Fact facts and keep your reasoning.

Trial F has height 6 cm and verified travel distance exactly 4 units. Apply Ramp1 and the **outcome rule**. A partner says exactly 4 is near because far means greater than 4. Use the stated rule to resolve the disagreement.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Keep incomplete evidence visible

> Keep incomplete evidence visible. Apply the supplied Prediction Is Not Fact facts and keep your reasoning.

D predicts near but has no distance record. E predicts far but its two distance records say 3 and 4 units.

State each comparison status, request useful **evidence**, and explain why neither guessing nor a new trial can establish the old trial's outcome.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Transfer the check to drying predictions

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Transfer the check to drying predictions. Apply the supplied Prediction Is Not Fact facts and keep your reasoning.

In a fictional paper test, Dry1 predicts slow for a moisture score of at least 7, quick otherwise. Allowed scores are whole numbers 1 through 10.

The observed outcome is slow if drying time is greater than 10 minutes, quick otherwise. Records are P: score 6, time 12; Q: score 7, time 10; R: score 9, time 14; S: score 5, time missing.

Record all **prediction**s, observed categories and comparison statuses. Summarise matches out of assessed cases, name the **unassessed** count, and limit the conclusion.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Correct a report without erasing evidence

> Correct a report without erasing evidence. Apply the supplied Prediction Is Not Fact facts and keep your reasoning.

After seeing B's observed near category, a learner replaces Ramp1's far **prediction** with near. They call the new entry a corrected **observation** and report a perfect result for A through C.

Explain the two record-keeping errors and write a defensible replacement report.

> **My reasoning and evidence**

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Prediction Is Not Fact

> A fictional drying trial uses a fixed rule and later measures what actually happened. Explain the labelled step before checking the changed condition.

A fictional drying trial uses a fixed rule and later measures what actually happened.

Model D predicts slow when moisture score is at least 5; quick otherwise.

Observation policy: slow if drying takes above 8 minutes; quick otherwise. Trial T has score 4 and measured time 9 minutes.

AI ACADEMY / AI-1/17

Prediction Is Not Fact

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.

1. Prediction input

T has moisture score 4.

2. Predict

4 is below 5, so D predicts quick.

3. Observe later

Measured time is 9 minutes; above 8 means observed slow.

4. Compare evidence

Prediction quick disagrees with observation slow.

CHANGE ONE THING
Keep score 4 but measure exactly 8 minutes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which field changes the **observation** label, and which field changes the model **prediction**?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: In sorter trials, record predicted categories before collecting reference **evidence**, then retain disagreements for investigation.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

In sorter trials, record predicted categories before collecting reference **evidence**, then retain disagreements for investigation.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional drying trial uses a fixed rule and later measures what actually happened.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Prediction** quick disagrees with **observation** slow.

> 1. Predict

2. Compare **evidence**

3. Observe later

4. **Prediction input**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare **evidence Prediction** quick disagrees with **observation** slow. The moisture score generates a **prediction**; it is not the observed drying time.

The later measurement supplies separate **evidence** under the stated **observation** policy. Save both values and their units. Altering the recorded time to match D would destroy the evaluation rather than repair the model.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Keep score 4 but measure exactly 8 minutes. Which conclusion is supported by the supplied **fact**s?

1. The observed label becomes quick and agrees with the **prediction**.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those **fact**s.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The observed label becomes

quick and agrees with the **prediction**. Above 8 is strict: equality is quick under this policy.

Repair a missing building block

Review Boundaries and 'Not Sure' (ai-1/16). Goal: Identify a specific source of uncertainty and explain how it affects a **prediction**. Short practice: Which original block needs a better width before accepting or rejecting it?

What should the reviewer do?. Return to this week and explain what you changed.

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Trace 0 cm and explain why clarification is different from predicting short.. Return to this week and explain what you changed.

Review Repeating Efficiently (ai-1/11). Goal: Trace a loop by testing its condition before every repetition. Short practice: Trace a fresh start of 9. How many additions occur, and must the report equal 10?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

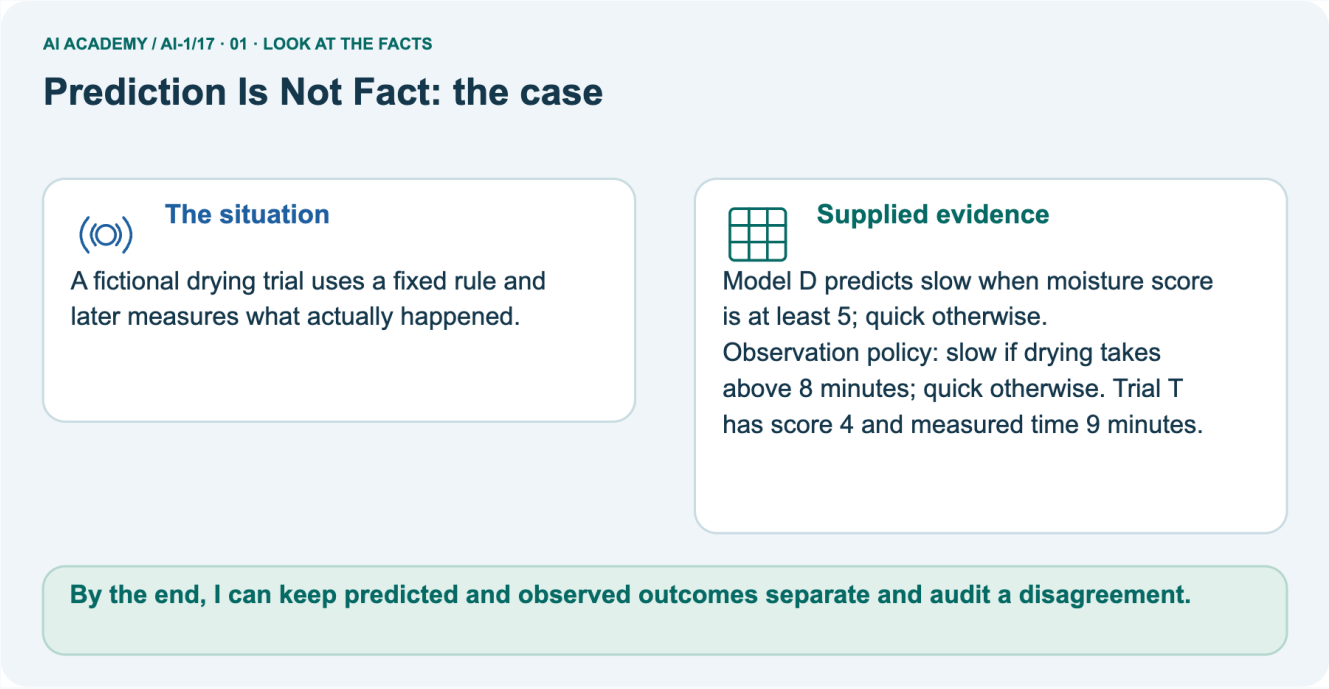
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Prediction Is Not Fact picture from its labels, then record a relevant limit.

01. Prediction Is Not Fact: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

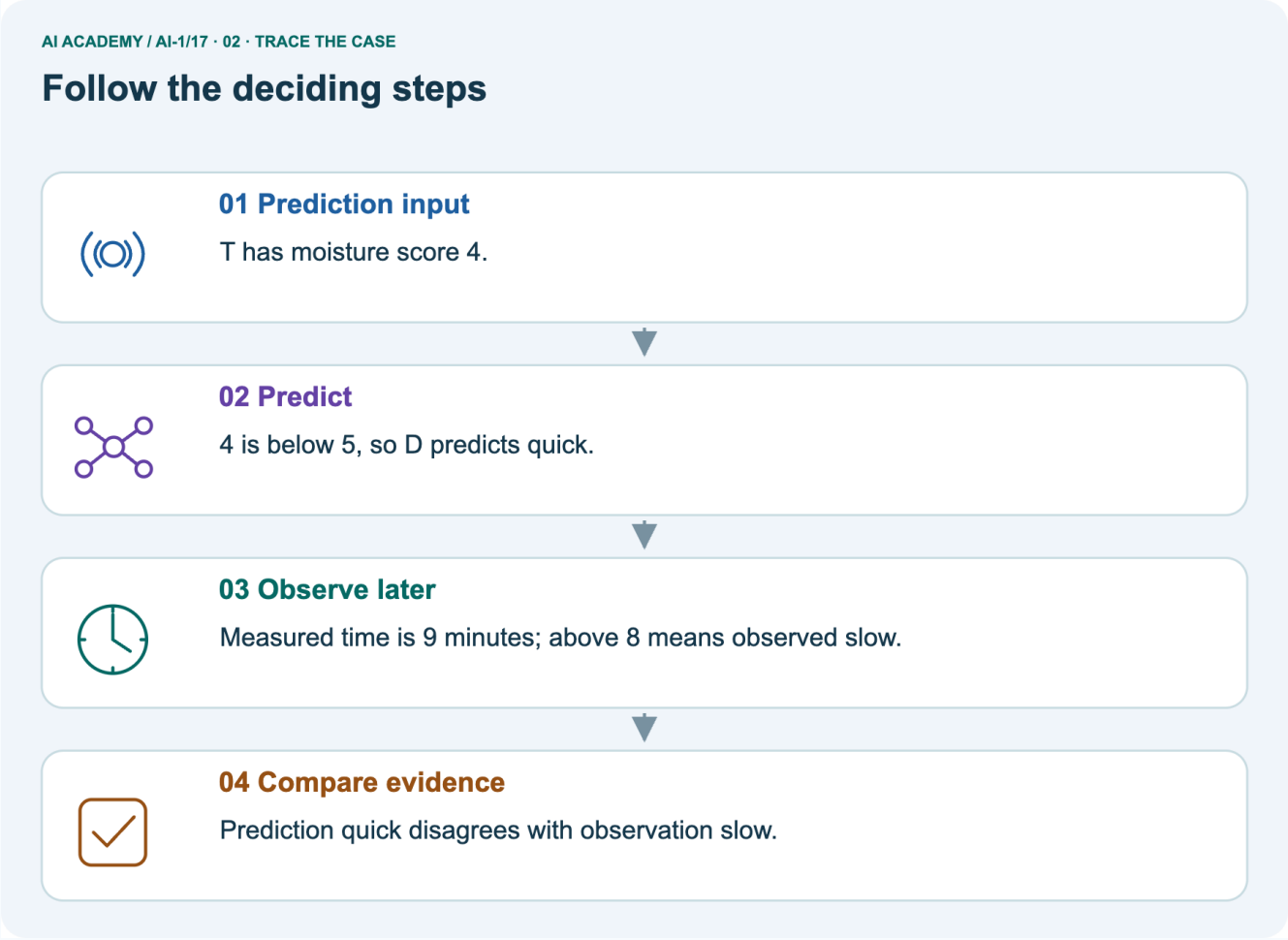
A fictional drying trial uses a fixed rule and later measures what actually happened. Supplied **fact**s: Model D predicts slow when moisture score is at least 5; quick otherwise.

Observation policy: slow if drying takes above 8 minutes; quick otherwise. Trial T has score 4 and measured time 9 minutes.

> Point to the given **fact** that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



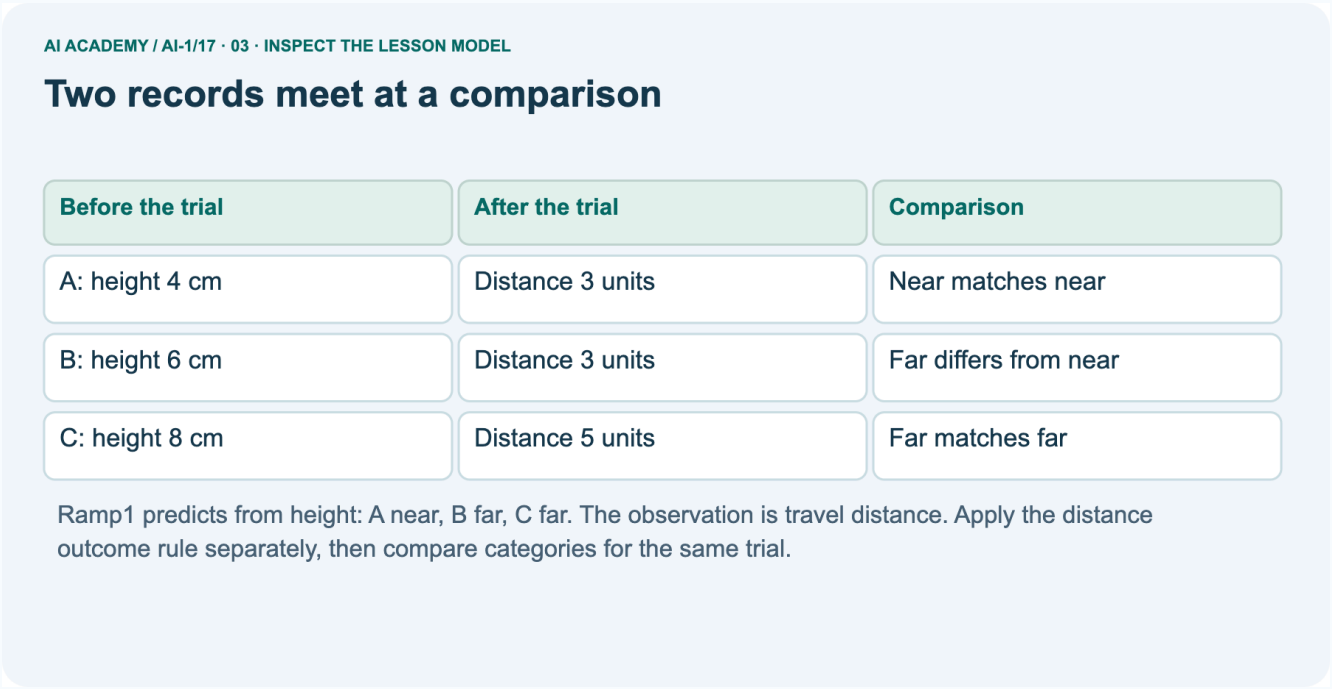
Follow the labelled picture with the matching explanation. Open the vector diagram

Prediction input: T has moisture score 4. → Predict: 4 is below 5, so D predicts quick. → Observe later: Measured time is 9 minutes; above 8 means observed slow. → Compare **evidence**: **Prediction** quick disagrees with **observation** slow.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two records meet at a comparison



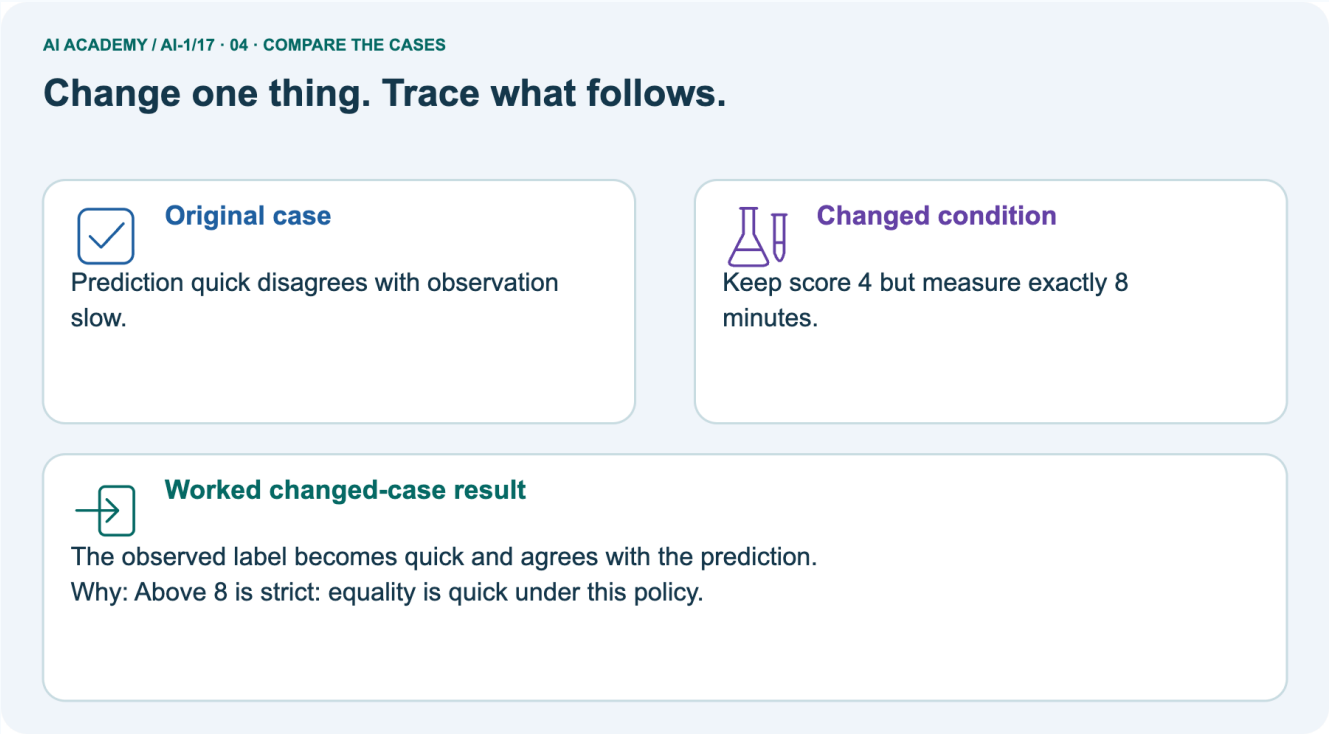
Follow the labelled picture with the matching explanation. Open the vector diagram

Two records meet at a comparison. A: height 4 cm | Distance 3 units | Near matches near. B: height 6 cm | Distance 3 units | Far differs from near. C: height 8 cm | Distance 5 units | Far matches far

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

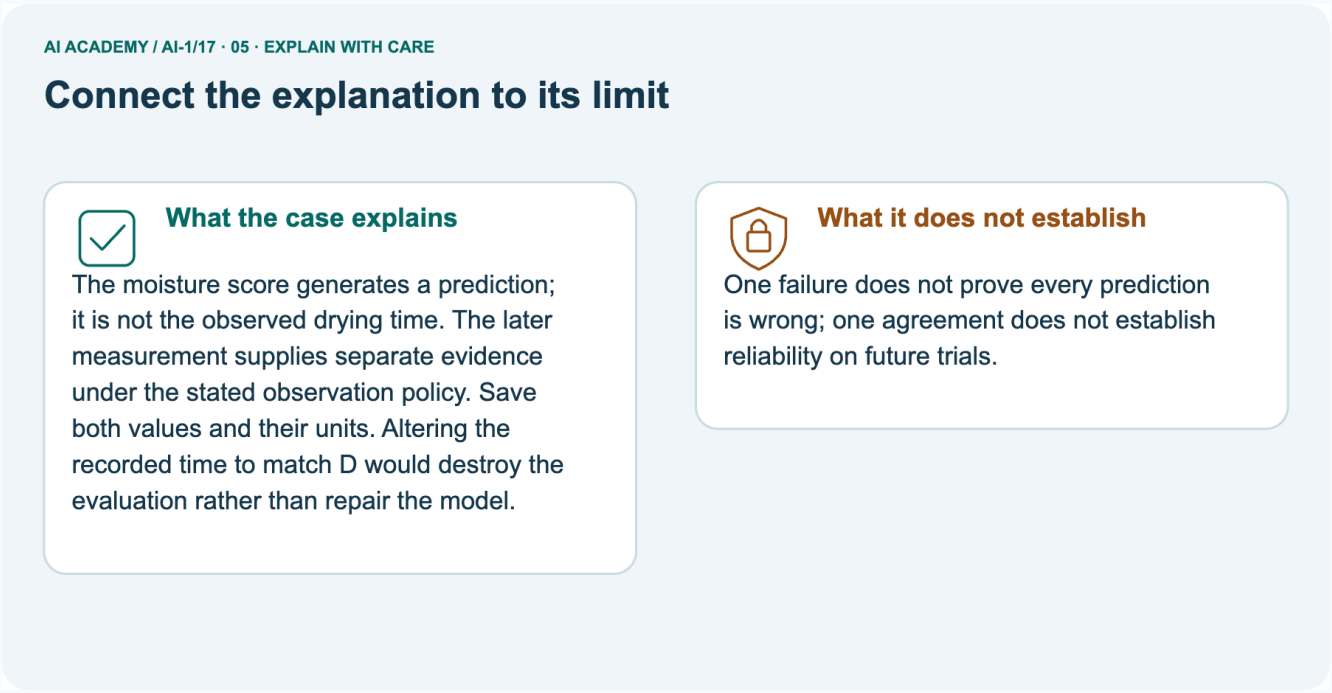
Original case: **Prediction** quick disagrees with **observation** slow. Changed condition: Keep score 4 but measure exactly 8 minutes. Expected result: The observed label becomes quick and agrees with the **prediction**.

Reason: Above 8 is strict: equality is quick under this policy.

Predict before checking the worked changed-case result. Name what changed and what the supplied **fact**s allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

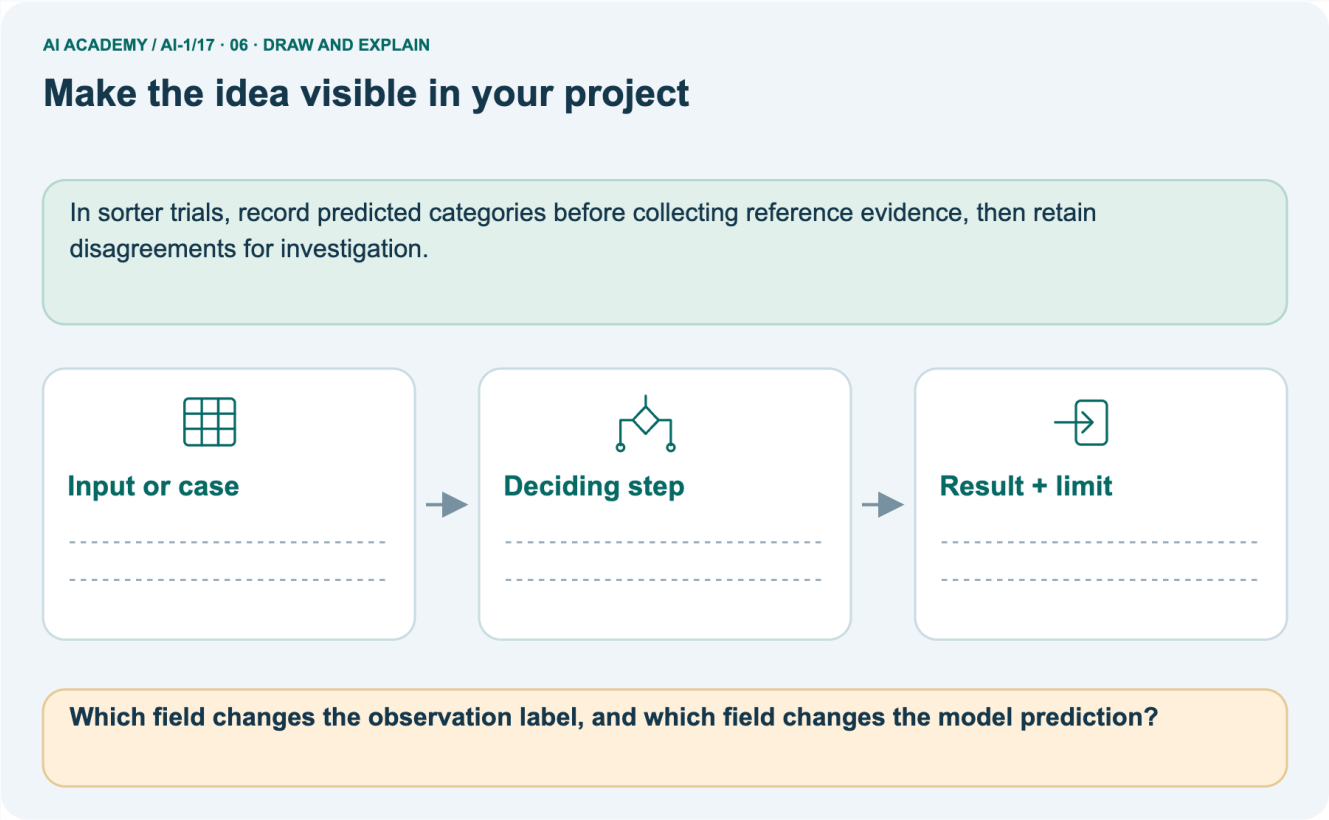
The moisture score generates a **prediction**; it is not the observed drying time. The later measurement supplies separate **evidence** under the stated **observation** policy. Save both values and their units.

Altering the recorded time to match D would destroy the evaluation rather than repair the model. Limit: One failure does not prove every **prediction** is wrong; one agreement does not establish reliability on future trials.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

In sorter trials, record predicted categories before collecting reference **evidence**, then retain disagreements for investigation. Draw your **input**, deciding step, result and limit.

Supported question: Which field changes the **observation** label, and which field changes the model **prediction**?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.